parthtejani / **periodic-subgraph-miner**

Implementation of an algorithm for periodic subgraph mining in dynamic networks

5 commits | 1 branch | 0 releases | 1 contributor

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parthtejani Fixed some formatting in README regarding horizontal lines — Latest commit `fa101ed` on 17 Dec 2012

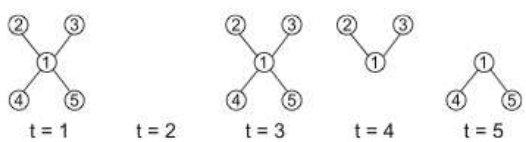
screenshots	Added screenshots	6 years ago
src/com/parthtejani/listminer	Added README and added fix to record correct time stamp for descriptors	6 years ago
.gitignore	Initial Commit	6 years ago
README.md	Fixed some formatting in README regarding horizontal lines	6 years ago

README.md

Periodic Subgraph Miner

This is an algorithm to mine periodic subgraphs in a dynamic network. The original algorithm is discussed in the paper "Periodic subgraph mining in dynamic networks" by Lahiri and Berger-Wolf, and the implementation is located here. The algorithm in this project is based on the paper "Speedup for a periodic subgraph miner" by Apostolico, Barbares, and Pizzi.

##Purpose The goal of the algorithm is to capture periodic and frequent interactions in a dynamic network. A dynamic network can be represented as a series of graphs at different time steps.



In this network, there are 4 periodic subgraphs with a minimum count of 2:

```
1. (1,2,3,4,5,6,7,8,9)  Period: 2       Start Time: 1   Count: 2
2. (1,4,5,8,9)                  Period: 2       Start Time: 1   Count: 3
3. (1,2,3,6,7)                  Period: 3       Start Time: 1   Count: 2
4. (1,2,3,6,7)                  Period: 1       Start Time: 3   Count: 2
5. (1)                                  Period: 1       Start Time: 3   Count: 3

*Edge 1-2 = 6, Edge 1-3 = 7, Edge 1-4 = 8, Edge 1-5 = 9
```

Usage

Input Arguments

Required Parameters

```
1. An input file                -i
2. An output file               -o
```

Input File

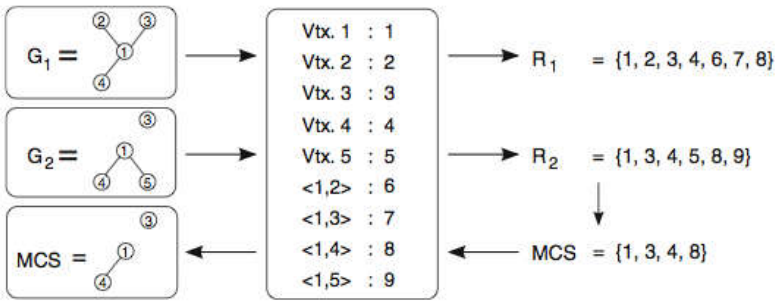
The input file contains lines with tokens separated by whitespace that represent the graph at a particular time step. The first token is the actual time (can be a second, a minute, a date, a year, etc). The rest of the tokens are a series of integers in ascending order that represent the graph at that particular timestamp.

Example input file text:

```
1s 1 2 3 4 5 6 7 8 9

3s 1 2 3 4 5 6 7 8 9
4s 1 2 3 6 7
5s 1 4 5 8 9
```

The integers representing the graph are uniquely mapped to vertexes and edges as shown below.



Output File

The output file contains data specific for grading this project. To view the actual periodic subgraphs, use the argument "-d" when calling the program to print the subgraphs to the console.

Example console output:

```
start 3 psup 2 p 1 m 0 [1 2 3 6 7]
start 1 psup 2 p 2 m 0 [1 2 3 4 5 6 7 8 9]
start 3 psup 3 p 1 m 0 [1]
start 1 psup 3 p 2 m 0 [1 4 5 8 9]
start 1 psup 2 p 3 m 0 [1 2 3 6 7]
```

start = start time, psup = support size/count, p = period, m = phase

Optional Parameters

```
1. Minimum support size (at least 2)           -s              default:3
2. Minimum period (at least 1)                         -Pmin   default:1
3. Maximum period (at least minimum period)    -Pmax   default:10
4. Smooth step (can only be 1)                         -smooth default:1
5. Turn subsumption off (flag only)                    -f
6. Print graphs to console (flag only)         -d
```

Time smoothing is not implemented because it was not required for this project.

Commands

Setup:

```
git clone https://github.com/parthtejani/periodic-subgraph-miner.git
cd periodic-subgraph-miner
mkdir bin
javac -d bin src/com/parthtejani/listminer/*.java
cd bin
```

Example Usage:

```
java com.parthtejani.listminer.ListMiner -i /some/input/file -o /some/output/file -s 3 -Pmin 1 -Pmax 10
```

You may need to change the heap size depending on the size of the input data, and the parameters used. The maximum heap size can be set with the "-Xmx" flag. -Xmx1024M sets the maximum heap size to 1024 MB.